

Focus of this homework:

- Fundamentals of forced vibration of damped SDOF vibratory systems under periodic and general excitations

Submit homework (in hard copies) by the beginning of the class. It is your responsibility to make sure that homework is submitted in good quality. Please organize your work neatly. Suggest to start each problem on a new page.

1) Rao 4.7

A slider-crank mechanism is used to impart motion to the base of a spring-mass-damper system, as shown in Fig. 4.38. Approximating the base motion $y(t)$ as a series of harmonic functions, find the response of the mass for the following parameters:

$m = 1 \text{ kg}$, $c = 10 \text{ N-s/m}$, $k = 100 \text{ N/m}$, $r = 10 \text{ cm}$, $\ell = 1 \text{ m}$, and $\omega = 100 \text{ rad/s}$.

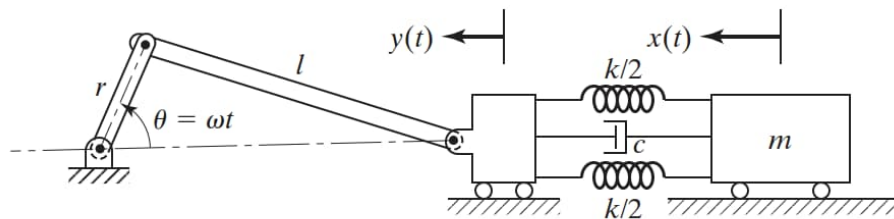
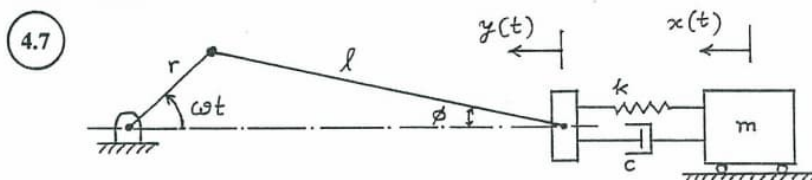


FIGURE 4.38 Input through slider crank mechanism

Modeling of the base excitation displacement is shown below.



Base motion is given by:

$$y(t) = r + \ell - r \cos \omega t - \ell \cos \phi = r + \ell - r \cos \omega t - \ell \sqrt{1 - \sin^2 \phi} \quad (1)$$

Using $\ell \sin \phi = r \sin \omega t$, Eq. (1) becomes

$$y(t) = r + \ell - r \cos \omega t - \ell \sqrt{1 - \frac{r^2}{\ell^2} \sin^2 \omega t} \quad (2)$$

Using the approximation:

$$\sqrt{1 - \frac{r^2}{\ell^2} \sin^2 \omega t} \approx 1 - \frac{r^2}{2\ell^2} \sin^2 \omega t \quad (3)$$

Eq. (2) can be expressed as

$$\begin{aligned} y(t) &= r + \ell - r \cos \omega t - \ell \left(1 - \frac{1}{2} \frac{r^2}{\ell^2} \sin^2 \omega t \right) \\ &= r - r \cos \omega t + \frac{\ell}{4} \left(\frac{r}{\ell} \right)^2 - \frac{\ell}{4} \left(\frac{r}{\ell} \right)^2 \cos 2\omega t \end{aligned} \quad (4)$$

2) Rao 4.8

The base of a spring-mass-damper system is subjected to the periodic displacement shown in Fig. 4.39. Determine the response of the mass using the principle of superposition. Plot (or sketch) the frequency components of the response up to the fifth order of the fundamental frequency.

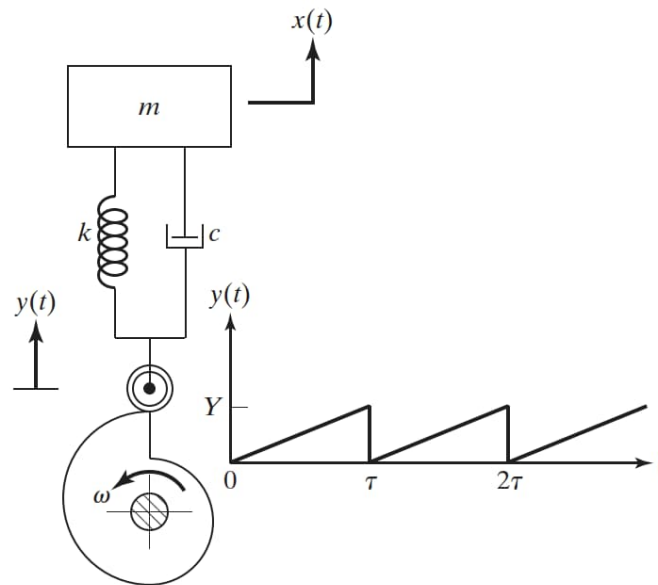


FIGURE 4.39 Periodic displacement input through cam.

3) Rao 4.39

4.39 The frame, anvil, and base of the forging hammer shown in Fig. 4.58(a) have a total mass of m . The support elastic pad has a stiffness of k . If the force applied by the hammer is given by Fig. 4.58(b), find the response of the anvil.

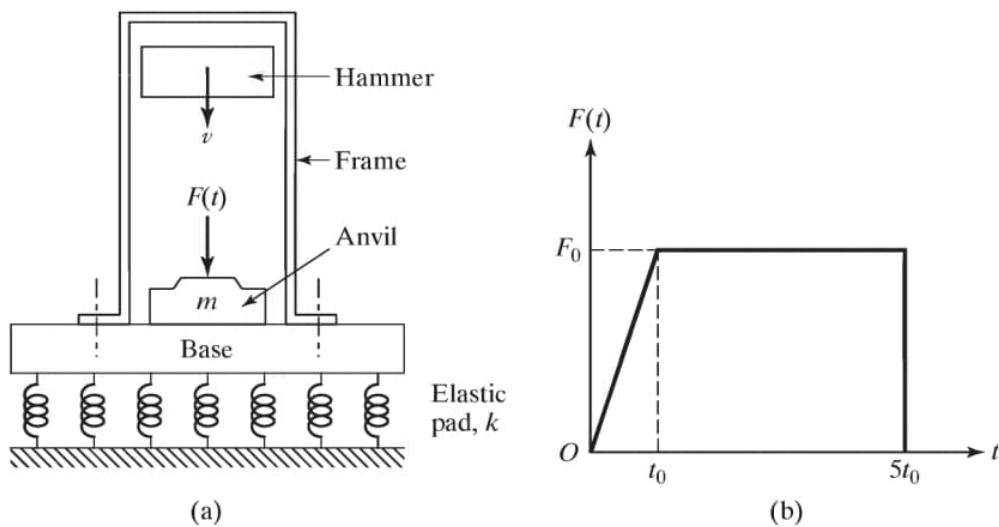


FIGURE 4.58